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T3

TSC Vision Cameras - Camera

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1 Document Revision History

Rev	Date	Author	Description
0.0.1	2024-11-01	Jan Svensson	First draft
0.0.2	2025-08-18	Jan Svensson	Change of highest ASIL C to ASIL B Clarification of FHTI and Nominal latency Added Receiver element as clarification Added TSR19 Camera ID

2 Scope and purpose of document

The TSC for Vision Cameras - Camera covers the following:

- Technical Safety Requirements derived from Functional Safety Concept and Safety Goals
- Applicable Architecture is T3
- Diagnostic Features
- Concerning the parts of belonging to the camera.

Derived from FSC Vision Cameras

3 TSC L1

3.1 Input Safety Goals

3.2 Input Justified Requirements

3.3 System design and requirements

3.3.1 First level technical safety requirements derived from functional safety requirement <FSR a>

3.3.2 First level technical safety requirements derived from functional safety requirement <FSR b>

3.4 Hardware metrics requirements

3.4.1 Hardware metrics requirements related to Safety Goal <SG1>

3.4.2 Hardware metrics requirements related to Safety Goal <SG2>

3.5 Diagnostic features

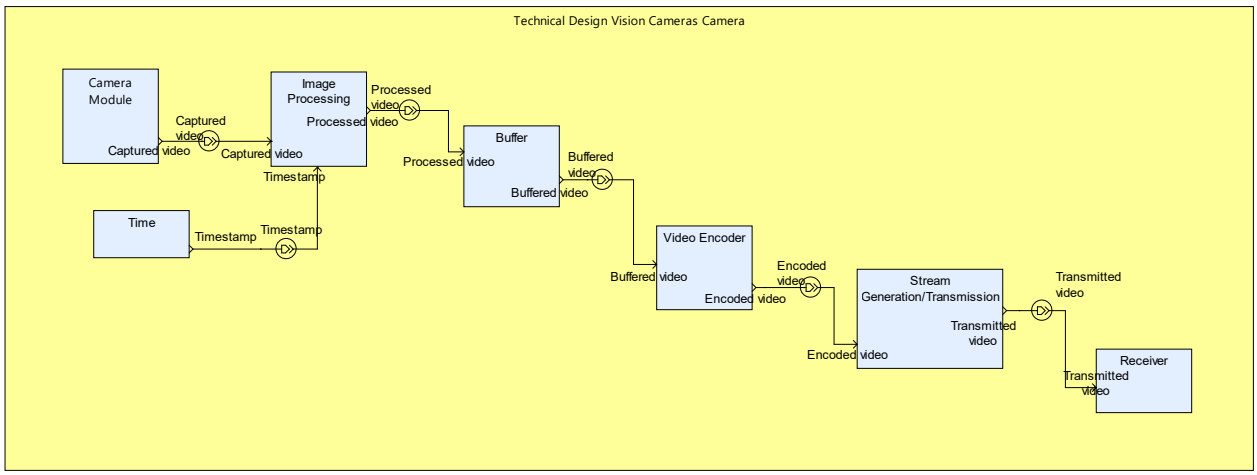
This involves the setting of Diagnostic Trouble Codes (DTCs) and possibly storage of additional information about the detected error and related circumstances for later read-out.

3.6 Technical safety requirements

The resulting technical safety requirements are listed below:

3.6.1 Technical Design Vision Cameras Camera

In the Technical Design the image pipeline is defined to consist of the elements Image Processing, Buffer, Video Encoder and Stream Generation/Transmission.



3.6.1.1 Architecture elements

3.6.1.1.1 Camera Module

Camera module element is responsible for image capture, convert incoming photons to a digital stream of pixels

3.6.1.1.2 Image Processing

Image processing element is responsible for any image processing that is applied before encoding video, like color & hue adjustment, fisheye restoration etc.

3.6.1.1.3 Time

Time element is responsible for keeping a syncronized local time.

3.6.1.1.4 Buffer

Buffer element represent the image buffer(s) in the image pipeline that could cause vulnerabilities due to storing old images. Ideally the buffer only stores small parts of the image before video encoding/compression.

3.6.1.1.5 Video Encoder

Video Encoder element applies compression to the video stream before sending

3.6.1.1.6 Stream Generation/Transmission

Stream Generation/transmission element takes the compressed video stream and send to receiver.

3.6.1.1.7 Receiver

Receiver element is the representation of reciever image pipeline elements and display element on TSC level, equals HMI Output element on FSC level

3.6.2 TSR1 Nominal transmission time

Requirement ID: F_Req-38258
Version: (1)
Status: CS Released
Changed: 2024-10-28 11:32

ASIL: **QM**()
*Allocated to element:***Camera Module***, in Architecture:***Technical Design Vision Cameras Camera***

Video frames shall be transmitted within 50 ms from actual traffic situation.

3.6.3 TSR2 Transmission time

Requirement ID: F_Req-38259
Version: (2)
Status: Frozen
Changed: 2025-08-18 11:29

ASIL: **B**()

Video frames shall be transmitted within FHTI ms from actual traffic situation alternatively provide means to detect transmission delays by receiver.

3.6.4 TSR3 Transmit video with integrity check

Requirement ID: F_Req-38260
Version: (1)
Status: CS Released
Changed: 2024-10-28 11:34

ASIL: **A**()
*Allocated to element:***Camera Module***, in Architecture:***Technical Design Vision Cameras Camera***

Video frames shall be transmitted such that it is possible to detect transmission integrity by receiver.

3.6.5 TSR4 Do not transmit unclear video

Requirement ID: F_Req-38261
Version: (1)
Status: CS Released
Changed: 2024-11-01 09:16

ASIL: **A**()
*Allocated to element:***Camera Module***, in Architecture:***Technical Design Vision Cameras Camera***

Video frames causing an unclear view (too dim, too low contrast, too much disturbances or noise, wrong resolution, flicker, too low framerate) shall not be transmitted.
Fulfilling this requirement means performing the activities needed to fulfil UN/ECE R46 from a camera module point of view.

3.6.6 TSR5 Do not transmit wrong FoV

Requirement ID: F_Req-38263

Version: (1)
Status: CS Released
Changed: 2024-10-28 12:08

ASIL: **B**()
*Allocated to element:***Camera Module***, in Architecture:***Technical Design Vision Cameras Camera***

The Camera module shall not transmit video frames with wrong field of view (e.g. cropped, rotated, flipped or partial images) unless requested by receiver.

3.6.7 TSR6 FHTI

Requirement ID: F_Req-38264
Version: (2)
Status: Frozen
Changed: 2025-08-18 11:27

ASIL: **B**()

Fault Handling Time Interval (FHTI) shall be 400 ms.
Note: FHTI shared with ECU (eg 200 + 200 ms) unless failures are proven independent from ECU.

3.6.8 TSR7 Camera module safe state

Requirement ID: F_Req-38265
Version: (2)
Status: Frozen
Changed: 2025-08-18 11:25

ASIL: **B**()

Failure to correctly transmit video frames shall trigger a transition to Camera module safe state, defined as halted image transmission and/or a notification to the receiver.

3.6.9 TSR8 Camera module safe state recovery

Requirement ID: F_Req-38266
Version: (1)
Status: CS Released
Changed: 2024-10-31 09:32

ASIL: **QM**()
*Allocated to element:***Camera Module***, in Architecture:***Technical Design Vision Cameras Camera***

In Camera module safe state, the camera shall monitor whether recovery from safe state can occur. If so, the camera shall transition back to normal operation.

3.6.10 TSR9 Receive and forward video

Requirement ID: F_Req-38267
Version: (2)
Status: Frozen
Changed: 2025-08-18 11:21

ASIL: **B**()

Video frames received by image processing in Image pipeline shall be correctly received, processed and forwarded to

Receiver within FHTI ms from actual traffic situation, else go to Camera ECU safe state.

3.6.11 TSR10 Do not forward unclear video

Requirement ID: F_Req-38268
Version: (1)
Status: CS Released
Changed: 2024-11-01 09:16

ASIL: **A**()
*Allocated to element:***Image Processing**, *in Architecture:***Technical Design Vision Cameras Camera**

Video frames causing an unclear view (too dim, too low contrast, too much disturbances or noise, wrong resolution, flicker, too low framerate) shall not be forwarded.
Fulfilling this requirement means performing the activities needed to fulfil UN/ECE R46 from a camera image processing point of view.

3.6.12 TSR11 Do not forward wrong FoV

Requirement ID: F_Req-38269
Version: (1)
Status: CS Released
Changed: 2024-10-31 09:33

ASIL: **B**()
*Allocated to element:***Image Processing**, *in Architecture:***Technical Design Vision Cameras Camera**

The Camera image processing shall not forward video frames with wrong field of view (eg cropped, rotated, flipped or partial images not requested by receiver), else go to Camera ECU safe state.

3.6.13 TSR12 FHTI ECU

Requirement ID: F_Req-38291
Version: (2)
Status: Frozen
Changed: 2025-08-18 11:22

ASIL: **B**()

Fault Handling Time Interval (FHTI) shall be 400 ms.
Note: FHTI shared with Camera module (eg 200 + 200 ms) unless failures are proven independent from Camera module.

3.6.14 TSR13 Camera ECU safe state

Requirement ID: F_Req-38292
Version: (2)
Status: Frozen
Changed: 2025-08-18 11:23

ASIL: **B**()

Failure to correctly forward video frames shall trigger a transition to Camera ECU safe state, defined as forwarding of blacked out images and/or a notification to the receiver.
Note: Safe state transitions are not mandatory as design choices that prevent the need to supervise could be applied.

3.6.15 TSR14 Camera ECU safe state recovery

Requirement ID: F_Req-38293
Version: (1)
Status: CS Released
Changed: 2024-10-31 09:34

ASIL: **QM**()
*Allocated to element:***Image Processing***, in Architecture:***Technical Design Vision Cameras Camera**

In Camera ECU safe state, the camera shall monitor whether recovery from safe state can occur. If so, the camera shall transition back to normal operation.

3.6.16 TSR15 Image timestamp insertion

Requirement ID: F_Req-38294
Version: (1)
Status: CS Released
Changed: 2024-10-29 12:41

ASIL: **QM**()
*Allocated to element:***Time***, in Architecture:***Technical Design Vision Cameras Camera**

A local time shall be used to generate an image timestamp to be inserted into each image.

3.6.17 TSR16 Synchronize time

Requirement ID: F_Req-38298
Version: (1)
Status: CS Released
Changed: 2024-10-29 12:42

ASIL: **QM**()
*Allocated to element:***Time***, in Architecture:***Technical Design Vision Cameras Camera**

A local time shall be synchronized with receiver time.

3.6.18 TSR17 Received video transmission time

Requirement ID: F_Req-38299
Version: (1)
Status: CS Released
Changed: 2024-10-29 12:44

ASIL: **QM**()
*Allocated to element:***Buffer***, in Architecture:***Technical Design Vision Cameras Camera**
*Allocated to element:***Video Encoder***, in Architecture:***Technical Design Vision Cameras Camera**
*Allocated to element:***Stream Generation/Transmission***, in Architecture:***Technical Design Vision Cameras Camera**

Video frames received by Image pipeline shall be transmitted to the receiver within 50 ms from reception.

3.6.19 TSR18 Transmit video with timestamp

Requirement ID: F_Req-38300
Version: (1)
Status: CS Released
Changed: 2024-10-29 12:45

ASIL: **QM**()
*Allocated to element:***Stream Generation/Transmission**, *in Architecture:***Technical Design Vision Cameras Camera**

Video frames shall be transmitted including an image timestamp such that it is possible to detect image delays by receiver.

3.6.20 TSR19 Transmit Camera ID

Requirement ID: F_Req-45283
Version: (1)
Status: Frozen
Changed: 2025-08-19 12:28

ASIL: **QM**()
*Allocated to element:***Stream Generation/Transmission**, *in Architecture:***Technical Design Vision Cameras Camera**

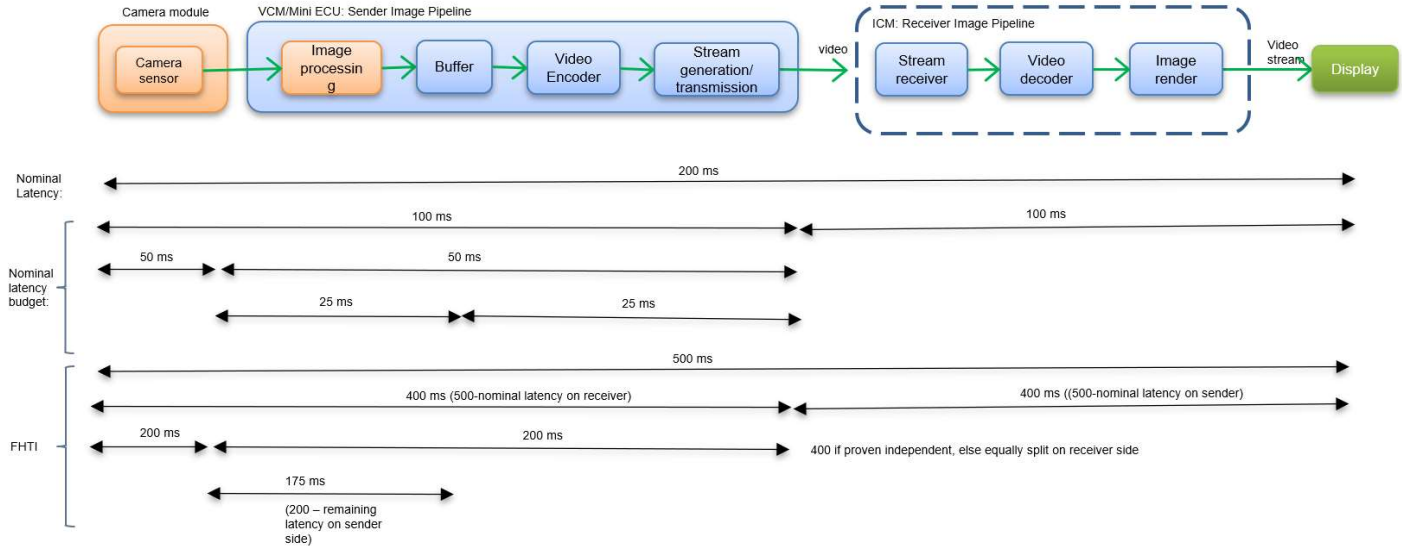
A camera ID originating from Receiver shall be embedded in image such that it is possible to detect a correct video stream among several streams by receiver.

4 Abbreviations

Abbreviation	Meaning	Description
ECE	Economic Commission for Europe	
ECU	Electronic Control Unit	
FHTI	Fault Handling Time Interval	Sum of fault detection time interval and the fault reaction time interval. Need to be < FTTI.
UN	United Nations	

5 Definitions

FHTI and Nominal latency breakdown: Clarification to requriements on FHTI and Nominal (QM) latency figures. The breakdown shows budgetary figures, meaning that they can be negotioated between elements as long as total FHTI and nominal latency is kept.



6 References

Reference	Document Number	Description	Referenced Document Version
UN/ECE R46		UN/ECE R46 regulation, https://unece.org/fileadmin/DAM/trans/main/wp29/wp29regs/2016/R046r6e.pdf	(1)
FSC Vision Cameras	50466650		(1)

7 History and ChangeLog

7.1 Requirements

7.1.1 Active requirements list

Requirement ID	Name	Version	Status
F_Req-38258	TSR1 Nominal transmission time	(1)	CS Released
F_Req-38259	TSR2 Transmission time	(2)	Frozen
F_Req-38260	TSR3 Transmit video with integrity check	(1)	CS Released
F_Req-38261	TSR4 Do not transmit unclear video	(1)	CS Released
F_Req-38263	TSR5 Do not transmit wrong FoV	(1)	CS Released
F_Req-38264	TSR6 FHTI	(2)	Frozen
F_Req-38265	TSR7 Camera module safe state	(2)	Frozen
F_Req-38266	TSR8 Camera module safe state recovery	(1)	CS Released
F_Req-38267	TSR9 Receive and forward video	(2)	Frozen
F_Req-38268	TSR10 Do not forward unclear video	(1)	CS Released
F_Req-38269	TSR11 Do not forward wrong FoV	(1)	CS Released
F_Req-38291	TSR12 FHTI ECU	(2)	Frozen
F_Req-38292	TSR13 Camera ECU safe state	(2)	Frozen
F_Req-38293	TSR14 Camera ECU safe state recovery	(1)	CS Released
F_Req-38294	TSR15 Image timestamp insertion	(1)	CS Released
F_Req-38298	TSR16 Synchronize time	(1)	CS Released
F_Req-38299	TSR17 Received video transmission time	(1)	CS Released
F_Req-38300	TSR18 Transmit video with timestamp	(1)	CS Released
F_Req-45283	TSR19 Transmit Camera ID	(1)	Frozen

7.1.2 New requirements

Requirement ID	Name	Version	Status
F_Req-45283	TSR19 Transmit Camera ID	(1)	Frozen

7.1.3 Updated requirements

Requirement ID	Name	Version	Status
F_Req-38259	TSR2 Transmission time	(2)	Frozen
F_Req-38264	TSR6 FHTI	(2)	Frozen
F_Req-38265	TSR7 Camera module safe state	(2)	Frozen
F_Req-38267	TSR9 Receive and forward video	(2)	Frozen
F_Req-38291	TSR12 FHTI ECU	(2)	Frozen
F_Req-38292	TSR13 Camera ECU safe state	(2)	Frozen

7.1.4 Removed requirements